



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SWASAN FILLING STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

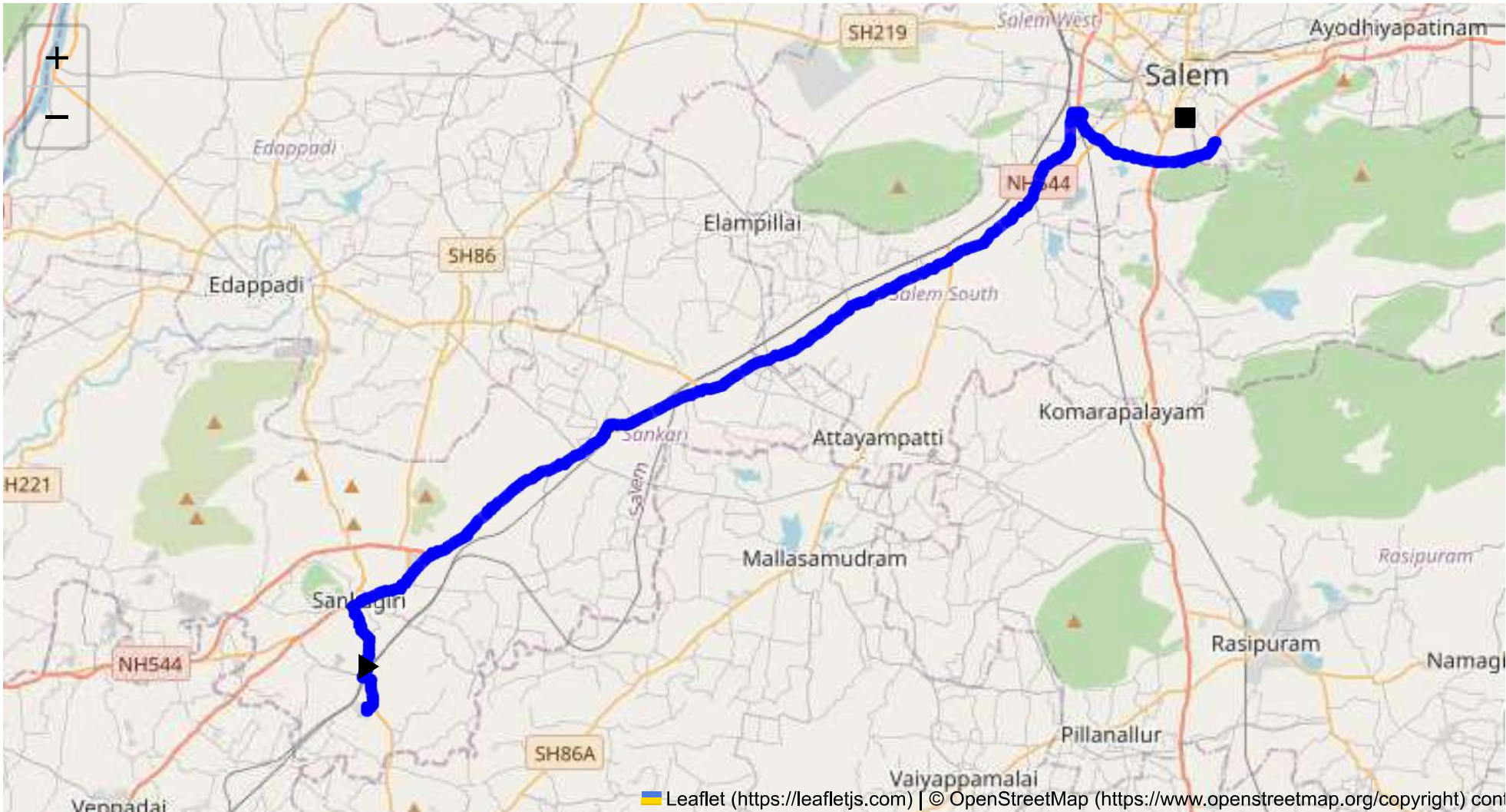
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 47.11 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.6252554, 78.15642347)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Seelanaickenpatti in Salem spans approximately 47.11 kilometers. It typically follows the NH-544, which is part of the Salem - Kochi corridor, eventually merging onto the Salem - Ulundurpettai Highway. The journey involves navigating through rural areas and transitioning into urban zones as you approach Salem.

Weather Conditions

- **Typical Weather:** The region typically experiences a tropical climate, with hot summers, mild winters, and a monsoon season from June to September.
- **Potential Hazards:** During the monsoon, the route might face heavy rainfall leading to water logging, reduced visibility, and increased risk of hydroplaning.

Traffic Patterns

- **Peak Hours:** Heavy traffic is usually encountered during morning and evening peak hours, roughly from 8 AM to 10 AM, and 5 PM to 8 PM.
- **Congestion-Prone Areas:** Expect congestion near major intersections and urban centers, particularly as you approach Salem.

Road Quality and Infrastructure

- **Road Quality:** The NH-544 is a well-maintained highway, but some sections may have potholes or uneven surfaces, particularly near Sangagiri and as you near urban areas in Salem.
- **Infrastructure:** Adequate signage, regular lane demarcations, and street lighting are generally present, but rural stretches may have less lighting.

Alternative Routes

- **Emergencies:** In case of road blocks or emergencies, an alternative route could be via the SH-86 which runs parallel to NH-544 but expect longer travel time.

Local Regulations on Hazardous Material

- **Regulations:** Transport of hazardous materials requires specific permits. Adherence to speed limits, signage, and nighttime driving restrictions for such vehicles must be strictly followed.

Historical Incidents

- While this route does not have a notorious history of major incidents, minor accidents involving heavy vehicles are not uncommon, often due to driver fatigue or adverse weather.

Environmental Considerations

- **Sensitive Areas:** The route passes through predominantly agricultural land. Care should be taken to prevent contamination of soil and water resources.

Communication Coverage

- **Potential Dead Zones:** Mobile network coverage is generally robust along NH-544 and near Salem, but rural areas may experience occasional disruptions.

Emergency Response Times

- Estimated Times:** Within urban areas near Salem, emergency services (ambulance, fire) can respond within 15-30 minutes. In rural areas and mid-route, response times could extend to 45-60 minutes.

Overall Risk Assessment

The primary risks involve navigating through congested urban areas, especially during peak hours, and potential weather-related challenges during the monsoon season. Proper planning and adherence to local regulations are critical to minimizing risks. Taking an alternate route during emergencies is advised, though it may result in increased travel time. Regular updates on weather and traffic conditions should be monitored to ensure safety. Despite some inherent risks, diligent attention to these factors can make the route manageable for transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	High	11.45004, 77.87206	15 KM/Hr
6	Turn	Medium	11.45028, 77.87202	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.47410, 77.86845	30 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.61736, 78.10588	30 KM/Hr
12	Turn	Medium	11.61753, 78.10608	30 KM/Hr
13	Blind Spot	Blind Spot	11.63219, 78.16753	10 KM/Hr
14	Turn	High	11.63211, 78.16762	15 KM/Hr
15	Blind Spot	Blind Spot	11.62746, 78.16194	10 KM/Hr
16	Blind Spot	Blind Spot	11.62732, 78.16198	10 KM/Hr
17	Turn	High	11.62564, 78.15656	15 KM/Hr
18	Turn	High	11.62522, 78.15646	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
23	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
25	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
26	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
27	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
28	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
29	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
30	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
31	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
32	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
33	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
24	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



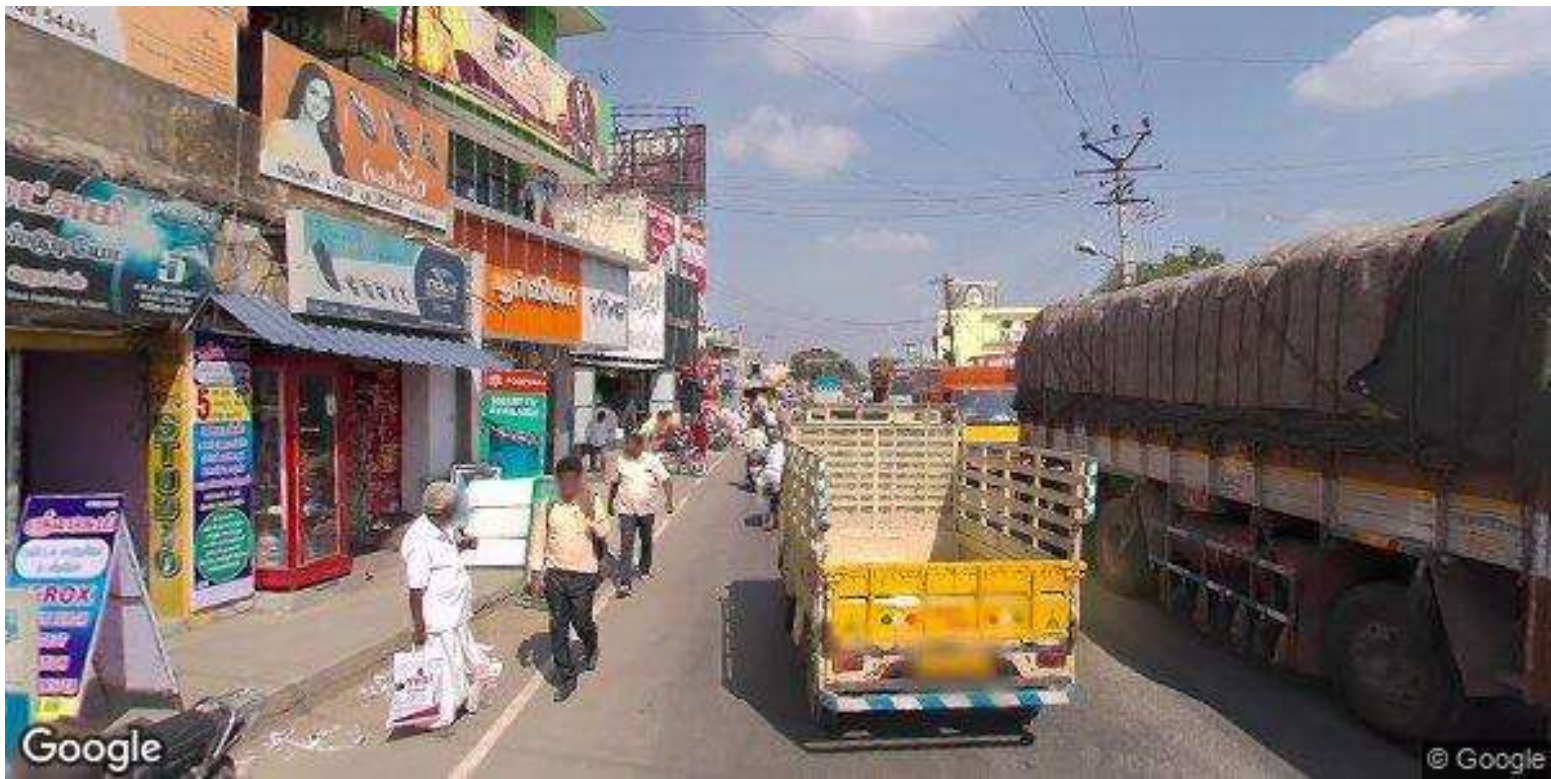
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45004, 77.87206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47410, 77.86845



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61753, 78.10608



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.63219, 78.16753



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63211, 78.16762



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62746, 78.16194



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62732, 78.16198



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62564, 78.15656



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62522, 78.15646